

RITESH VERMA

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CAREER OBJECTIVE:

Backend Developer with hands-on experience building RESTful APIs and scalable web applications using Python, Django, and Flask. Skilled in database design with MySQL/PostgreSQL, writing clean and testable code, and version control with Git. Additionally developed machine learning projects involving computer vision and fraud detection, applying strong analytical and problem-solving skills. Solid foundation in data structures, algorithms, and system design, with a quick-learning mindset and eagerness to contribute to a backend engineering team

EDUCATION:

Pranveer Singh Institute of Technology (PSIT), Kanpur

B.Tech, Computer Science & Engineering |2024-2028

Overall CGPA : 8.56/10 (up to 4th semester)

12th | CBSE/DAV Public School | 2023 – 85%

10th | CBSE/DAV Public School | 2021 – 88%

TECHNICAL SKILLS:

Languages: C, C++, Python, R, JavaScript, SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Web Development: HTML, CSS, JavaScript, React/Node.js

Tools & Platforms: Git, GitHub, VS Code, Eclipse, Docker, PyCharm, Jupyter Notebook,

Areas of Interest: Software Development, Problem Solving, Machine Learning, Web Development

PROJECTS:

ClearSight | Python, PyTorch, U-Net, Pix2Pix GAN, OpenCV, GDAL

- Built an AI-based satellite image restoration system to detect and remove cloud coverage from LISS-IV imagery.
- Implemented U-Net-based cloud segmentation to generate pixel-level cloud masks and identify affected regions.
- Developed a Pix2Pix GAN-based reconstruction pipeline to restore cloud-covered surface information and generate cloud-free imagery.

QuantumShield AI | React.js, Python, FastAPI, Machine Learning, Qiskit

- Built a full-stack financial transaction fraud detection platform using AI/ML and Quantum Machine Learning to identify suspicious transactions.
- Designed and implemented real-time fraud prediction, risk scoring, analytics dashboards, user authentication, and admin model management.
- Applied XGBoost, Random Forest, SHAP, QSVM, and Variational Quantum Classifiers to improve fraud detection, model interpretability, and quantum-vs-classical benchmarking.

Tech Stack: React.js, TypeScript, Tailwind CSS, Python, FastAPI, PostgreSQL, Scikit-learn, XGBoost, SHAP, Qiskit, Redis, Docker

ACHIEVEMENTS:

- Solved 350+ problems on LeetCode and 200+ problems on HackerRank, strengthening Data Structures & Algorithms and problem-solving skills.
- Successfully submitted an idea and participated in the Bharatiya Antariksh Hackathon 2026, presented by ISRO.

CERTIFICATIONS:

- DevOps Fundamentals: Git, CI/CD & Docker Basics** — Udemy
- Data Science Bootcamp** — GeeksforGeeks
- Figma AI Workflow: From Vision to Version** — Udemy
- Discover Your Business Priority** — Udemy